

Final Project Report

Parity: Citizen Grievance Dashboard

1. Introduction

Citizen grievances are a vital feedback mechanism for governance. However, when complaints are scattered across spreadsheets or reports, they become difficult to analyze and act upon. This project introduces the **Parity Dashboard**, a data-driven solution that transforms raw grievance records into interactive, visual insights for better decision-making.

2. Problem Statement

Citizens raise thousands of complaints every year, covering issues such as sanitation, water supply, and infrastructure. These grievances often remain unstructured, making it challenging for authorities to:

- Identify recurring issues.
- Prioritize wards/categories needing urgent attention.
- Track complaint trends over time.
- Listen to the “voice of the citizen” in a meaningful way.

3. Solution

The **Parity Dashboard** addresses these challenges by:

- Centralizing grievance data into a single platform.
- Providing interactive filters for ward, category, and date.
- Visualizing complaint trends with charts and graphs.
- Highlighting frequent issues through word clouds.
- Ranking wards/categories in a leaderboard for accountability.

This makes governance more **transparent, responsive, and citizen-centric**.

4. Tools Used

- **Python** – Core programming language.
- **Pandas** – Data cleaning and analysis.
- **Streamlit** – Interactive dashboard framework.
- **Matplotlib / Plotly** – Charts and graphs.

- **WordCloud** – Text visualization.

5. Features

- **KPIs:** Total complaints, wards covered, categories.
- **Leaderboard:** Top wards/categories with most complaints.
- **Trend Analysis:** Complaints over time.
- **Filters:** Ward, category, date range.

Word Cloud: Citizen voice visualization.

- **Download Option:** Export filtered complaints as CSV.
- **Optional Add-ons:** Sentiment analysis, map visualization.

6. Instructions to Run

1. Install dependencies:

```
bash
```

```
pip install streamlit pandas matplotlib wordcloud plotly
```

2. Place your CSV file (2025-grievances.csv or clean_grievances.csv) in the project folder.

3. Run the dashboard:

```
bash
```

```
streamlit run parity_dashboard.py
```

4. Open the link in your browser (default: <http://localhost:8501>).
5. Interact with filters, charts, leaderboard, and word cloud.

7. Conclusion

The **Parity Dashboard** bridges the gap between citizen grievances and actionable governance. By making complaints visible, measurable, and interactive, it empowers decision-makers to prioritize issues, allocate resources, and improve public services. This project demonstrates how technology can amplify citizen voices and foster accountability in governance.